

Combustion Chemistry

Outline *After looking at the thermodynamics and kinetics of combustion-relevant chemical processes, the current handout digs deeper on the chemistry of combustion in terms of reaction mechanisms. As seen in the previous handout, a simple global reaction is usually a result of hundreds and thousands of elementary reactions, each involving their own kinetics and thermodynamic analysis. Most of the time, a large number of chemical species are involved even in combustion of simple hydrocarbons. This usually renders simple analytical solutions impossible for these systems and a computational model is usually adopted. The interplay of thermo-chemistry and chemical kinetics can be easily seen in these real world application of combustion related chemistry and can lead to some interesting phenomenons. In this handout, we will take a look at some of these reaction mechanisms which are important in the study of combustion systems.*

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1 Hydrogen-Oxygen Reaction

The reaction between hydrogen and oxygen is a good example of a multi-component kinetic system. To describe the system properly, we should consider eight major species and at least 16 reactions. The overall reaction is given by:



This reaction is exothermic; but mixtures of gaseous hydrogen and oxygen are quite stable at atmospheric conditions. Any conceivable direct reaction between the two gases is zero. The reaction half-time at atmospheric conditions has been estimated to be much larger than the age of the universe. However, if the reaction is initiated by some free-radical species, then the reaction proceeds very rapidly and violently. The radicals are typically H and O atoms produced from the dissociation of H_2 and O_2 , respectively. We will consider a simplified mechanism that represents the gross features of $\text{H}_2 + \text{O}_2$ at low P .

- **Initiation:**



- **Chain branching:**



- **Chain propagation:**



- **Chain termination:**



Initiation step is the dissociation of some amount of molecular species by a spark, flame, electric discharge, or some other means. Reaction (H.2) is endothermic by 0.7 eV (about 70 kJ/mol) and progresses slowly. Reactions (H.3) and (H.4) are endothermic by 0.1 eV, and these reactions are relatively fast. The OH and O radicals are therefore rapidly consumed, and the principal chain carrier is H atoms. We will find the rate equation for the free-radical density n^* , which is taken to be the same as that for [H] atoms:

$$\frac{dn^*}{dt} = \frac{d[\text{H}]}{dt} = k_o[\text{H}_2] - k_2[\text{H}][\text{O}_2] + k_3[\text{O}][\text{H}_2] + k_1[\text{OH}][\text{H}_2] - k_4[\text{H}] - k_5[\text{H}][\text{O}_2][\text{M}] \quad (2)$$

For other free-radical species:

$$\frac{d[\text{OH}]}{dt} = k_2[\text{H}][\text{O}_2] + k_3[\text{O}][\text{H}_2] - k_1[\text{OH}][\text{H}_2] \quad (3)$$

$$\frac{d[\text{O}]}{dt} = k_o[\text{O}_2] + k_2[\text{H}][\text{O}_2] - k_3[\text{O}][\text{H}_2] \quad (4)$$

Since $[\text{O}]$ and $[\text{OH}]$ are both much lower than $[\text{H}]$, we can assume that both of these species are at steady-state:

$$\frac{d[\text{OH}]}{dt} = \frac{d[\text{O}]}{dt} = 0 \quad (5)$$

that yields

$$[\text{O}]_{\text{ss}} = \frac{k_o[\text{O}_2] + k_2[\text{H}][\text{O}_2]}{k_3[\text{H}_2]} \quad (6)$$

and for $[\text{OH}]$,

$$[\text{OH}]_{\text{ss}} = \frac{k_2[\text{H}][\text{O}_2] + k_3[\text{O}][\text{H}_2]}{k_1[\text{H}_2]} \quad (7)$$

Substituting for $[\text{O}]_{\text{ss}}$,

$$[\text{OH}]_{\text{ss}} = \frac{2k_2[\text{H}][\text{O}_2] + k_o[\text{O}_2]}{k_1[\text{H}_2]} \quad (8)$$

We substitute the steady-state values of $[\text{OH}]$ and $[\text{O}]$ radicals in the rate equation for free-radical density

$$\begin{aligned} \frac{dn^*}{dt} &= k_o[\text{H}_2] - k_2[\text{H}][\text{O}_2] + k_3 \frac{k_2[\text{H}][\text{O}_2] + k_o[\text{O}_2]}{k_3[\text{H}_2]} [\text{H}_2] \\ &\quad + k_1[\text{H}_2] \frac{2k_2[\text{H}][\text{O}_2] + k_o[\text{O}_2]}{k_1[\text{H}_2]} - k_4[\text{H}] - k_5[\text{H}][\text{O}_2][\text{M}] \\ &= \underbrace{k_o([\text{H}_2] + 2[\text{O}_2])}_{w_o} + \underbrace{(2k_2[\text{O}_2] - k_4 - k_5[\text{O}_2][\text{M}])}_{f} n^* \quad (9) \end{aligned}$$

so that

$$\frac{dn^*}{dt} = w_o + (f - g)n^* \quad (10)$$

Here, f represents the contribution from chain branching and g from chain termination. Two classes of solutions are possible:

- **$g > f$: termination exceeds branching.** This solution implies that:

$$k_4 > (2k_2 - k_5[\text{M}])[\text{O}_2] \quad (11)$$

so that $g > f$ is assured at sufficiently low O_2 pressures. The solution is then

$$n^* = \frac{w_o}{g - f} \{1 - e^{-(g-f)t}\} \quad (12)$$

At short times, n^* increases almost linearly with slope $\sim w_o t$ and reaches a steady-state

$$n_{ss}^* = w_o / (g - f) \quad (13)$$

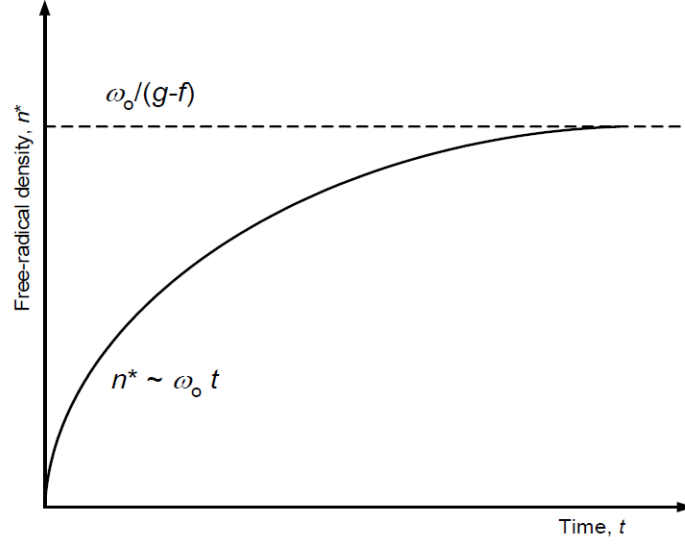


Figure 1: Temporal evolution of free radical density at sufficiently low O₂ pressures.

- $g < f$: **branching exceeds termination.** At higher O₂ pressures $g < f$, then

$$n^* = \frac{w_o}{f - g} \{e^{(f-g)t} - 1\} \quad (14)$$

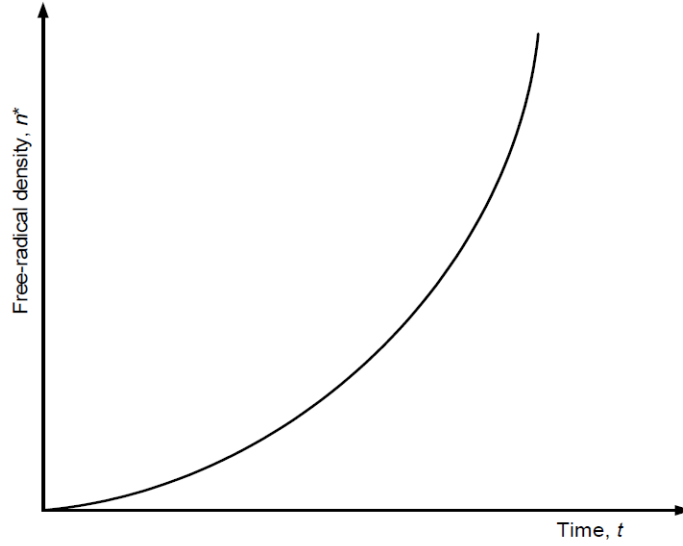


Figure 2: Temporal evolution of free radical density at higher O₂ pressures

In this case, the free-radical concentration increases exponentially, and, since the overall rate depends on the radical concentration, the reaction velocity increases rapidly. This is usually termed as an explosion.

The hydrogen-oxygen reaction behaves quite differently in different pressure regimes. It is possible to construct an explosion boundary for the hydrogen-oxygen reaction as a function of temperature and pressure. Other reactions, in addition to those we considered above, are important and including the following:

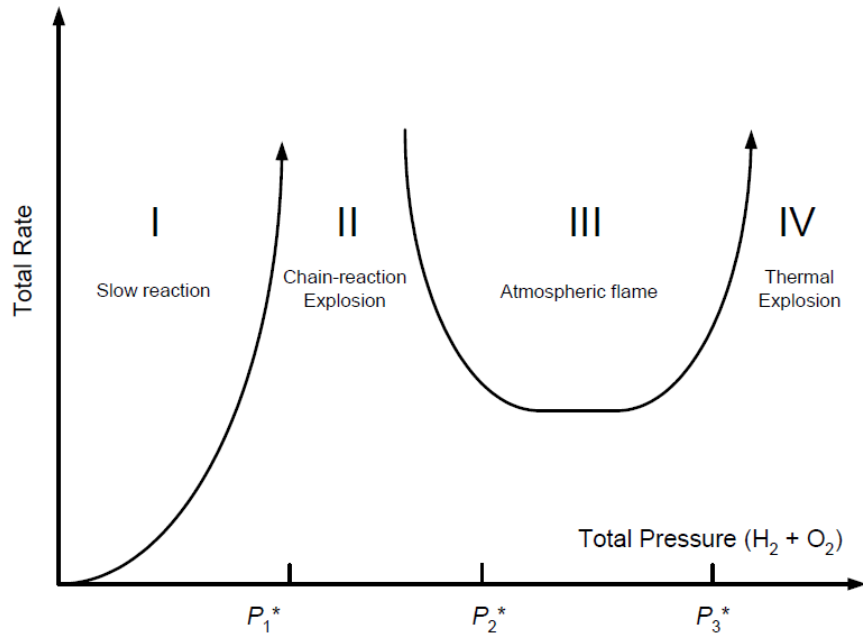


Figure 3: Classification of Hydrogen-Oxygen overall reaction with changing pressures.

- **Region I:** The reaction is wall-recombination limited, and proceeds to a steady-state.
- **Region II:** f begins to exceed g , the branched-chain reaction takes over and explosion ensues.
- **Region III:** As the pressure is further increased, the explosion is quenched and another

regime of steady-state is encountered. In III, kinetics are dominated by relatively unreactive HO_2 (hydroperoxyl) radicals.

- **Region IV:** At a still high pressure, the amount of heat liberated in the exothermic steps of the mechanism becomes larger than can be dissipated by conduction and other thermal transport processes, and the temperature rises. This, in turn increases the rates of initiation and provides heat for endothermic chain-branching reactions leading to more heat release, resulting in a thermal explosion.

Experimentally observed ($P - T$) boundaries of the $\text{H}_2\text{-O}_2$ reaction in a closed vessel is shown below.

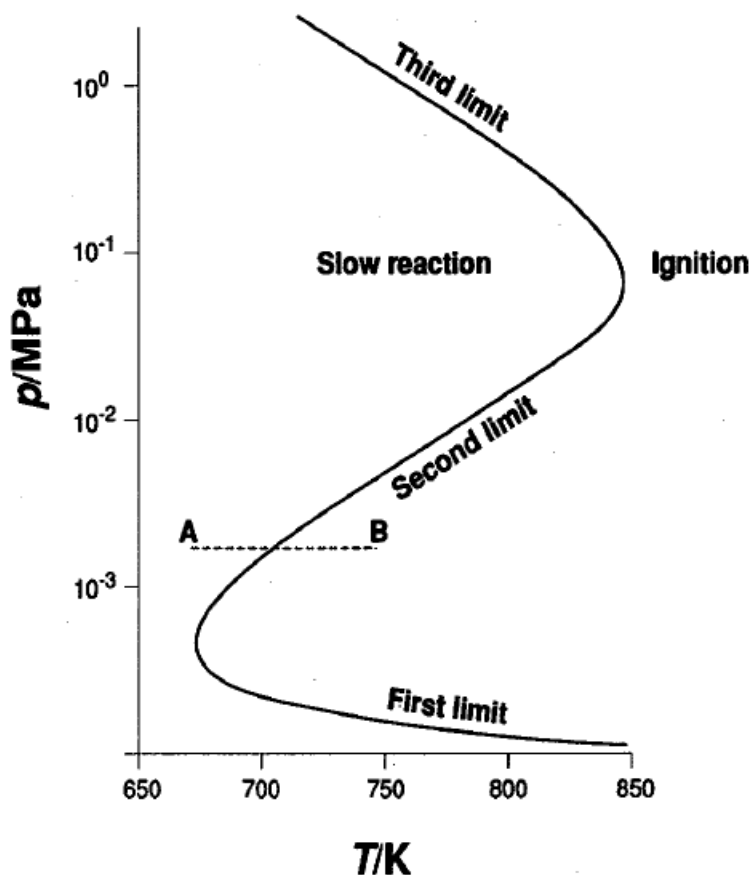


Figure 4: Explosion limits for a stoichiometric hydrogen–oxygen mixture in a spherical vessel.

2 Carbon Monoxide Oxidation

Oxidation of CO is important in hydrocarbon combustion. From a very simplistic point-of-view, hydrocarbon combustion (related to C content) can be characterized as a two-step process: breakdown of fuel to CO, and oxidation of CO to CO_2 . CO oxidation is extremely slow in the absence of small amounts of H_2 or H_2O . This is because the CO oxidation step involving the hydroxyl radical is

much faster than the steps involving O_2 and O . If H_2O is the primary hydrogen-containing species, the CO oxidation can be described by:



CO.1 is slow and therefore, does not contribute much to CO_2 formation, but serves as a chain initiation reaction. CO.3 is the actual CO oxidation step and also the chain-propagation step producing H atoms. CO.2 and CO.4 are chain-branching reactions producing OH, and OH and H, respectively. These radicals, then feed back to CO.2 and CO.3 causing further chain branching and oxidation respectively. CO.3 is the main component of this reaction scheme. On the other hand, if H_2 is present as the catalyst instead of water, then following steps are involved:



In the presence of H_2 , the entire H_2 - O_2 reaction system should also be included to describe CO oxidation.

3 Oxidation of Higher Paraffins

Alkanes, or paraffins, are saturated, straight-chain or branched-chain, single-bonded hydrocarbons with the general molecular formula C_nH_{2n+2} . Since methane (and ethane) display some unique characteristics not common with higher alkanes, we will be looking at paraffins with $n > 2$. The oxidation of alkanes can be characterized by three sequential processes, given below

- Fuel is attacked by O and H; breaks down to H_2 and olefins (double-bonded straight hydrocarbons). H_2 oxidizes to H_2O .
- Unsaturated olefins form CO and H_2 . Almost all H_2 converts to water.
- CO burns to CO_2 releasing almost all of the heat associated with combustion as shown in CO.3

These three processes can be further detailed as (with the example of propane, C_3H_8):

- **Step#1** : A C-C bond is broken in the original fuel molecule. A C-C bond is weaker than an H-C bond.



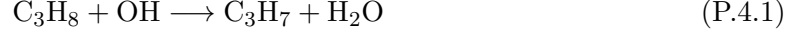
- **Step#2** : Two resulting hydrocarbon radicals break down further to olefins: H-atom abstraction.



- **Step#3** : H atoms from Step#2 starts a radical pool:



- **Step#4** : With the development of a radical pool, attack on the fuel molecule intensifies:



- **Step#5** : Hydrocarbon radicals decay to olefins and H atoms via H-atom abstraction.



This process obeys the β -scission rule, which states that C-C or C-H bond broken will be the one that is one place removed from the radical site.

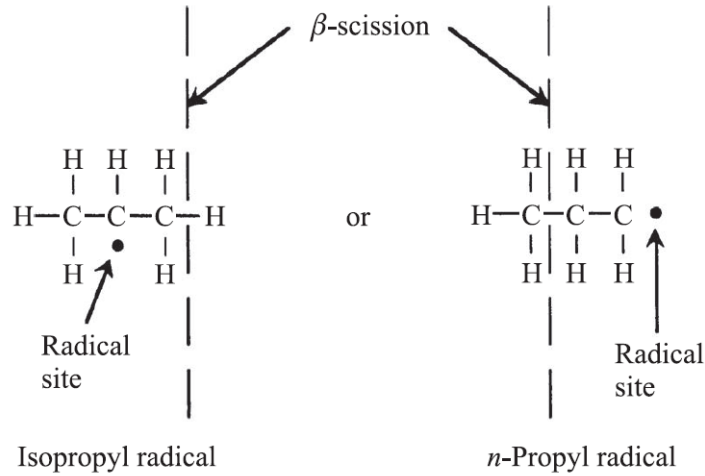
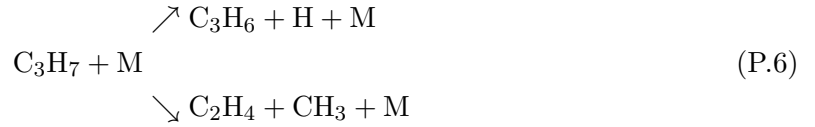


Figure 5: The β -scission rule applied to bond breaking in C_3H_7 radical

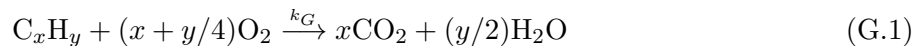
- **Step#6** : Oxidation of olefins created in Steps#2 and 5 by O that produces formyl radicals (HCO) and formaldehyde (H_2CO).



- **Step#7** : Methyl radicals (CH_3), formaldehyde (H_2CO), and methylene (CH_2) oxidize.
- **Step#8** : Carbon monoxide oxidizes following the CO mechanism discussed previously.

4 Global and Quasi-Global Reactions

The sequential nature of the processes I–III above has led to empirical global models that capture the overall behavior in a sequence of global or quasi-global steps. Global models, by definition do not capture all the features of hydrocarbon combustion, but they may be useful in simple engineering approximations as long as their limitations are recognized. Consider this single step global reaction



The rate expression for this reaction is written as:

$$\frac{d[\text{C}_x\text{H}_y]}{dt} = -A \exp[-E_a/(R_u T)] [\text{C}_x\text{H}_y]^m [\text{O}_2]^n \quad (\text{G.2})$$

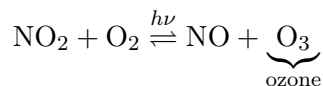
where the values of A , E_a , m and n are empirically derived for each combination of x and y to best match the experimental observations. Similar to a one step mechanism but with some more details and accuracy, an example of a multi-step global mechanism is :



which assumes that the intermediate hydrocarbon is ethylene (C_2H_4). Herein again, the rate expressions for each of these reactions is empirically adjusted to best match the experimental trends of the global reaction.

5 Oxides of Nitrogen Formation

Combustion products contain NO at levels of several hundred to several thousand parts per million (ppm) and NO_2 levels in tens of ppm. In the atmosphere, in the presence of ultraviolet sunlight, an equilibrium is established:



Presence of certain hydrocarbons (e.g., unburned hydrocarbons from combustion, methane from various sources) slowly unbalances the above reaction. NO contributes to destruction of ozone in stratosphere, production of ground level ozone and is involved in photochemical smog and haze. The main sources of nitrogen oxide, NO_x , emissions from combustion are:

- *Thermal* NO: oxidation of molecular nitrogen in the postflame zone.
- *Prompt* NO: formation of NO in the flame zone (*Fenimore* mechanism).
- *N₂O-intermediate* mechanism.
- *Fuel* NO: oxidation of nitrogen-containing compounds in the fuel.

Relative importance of these three are dependent on the operating conditions and fuel. In most practical combustion devices the thermal NO is the main source. The basic mechanism for thermal NO production is given by six reactions (forward and backward) known as **extended Zeldovich mechanism**:



The contribution of reaction N.3 is small for lean mixtures, but for rich mixtures it should be considered. Forward reaction N.1 controls the system, but it is slow at low temperatures (high activation energy). Thus it is effective in post-flame zone where temperature is high and residence time is available. Concentrations of 1000 to 4000 ppm are typically observed in uncontrolled combustion systems. From reactions 1-3, the rate of formation of thermal NO can be calculated:

$$\frac{d[\text{NO}]}{dt} = k_{1f}[\text{O}][\text{N}_2] - k_{1r}[\text{NO}][\text{N}] + k_{2f}[\text{N}][\text{O}_2] - k_{2r}[\text{NO}][\text{O}] + k_{3f}[\text{N}][\text{OH}] - k_{3r}[\text{NO}][\text{H}] \quad (15)$$

To calculate the NO formation rate, we need the concentrations of O, N, OH, and H. In detailed calculations, these are computed using detailed kinetic mechanisms for the fuel used. For very approximate calculations, these may be assumed to be in chemical equilibrium. At moderately high temperatures N does not stay at thermodynamic equilibrium. A better approximation could be to assume N to be at steady-state. From reactions N.1,N.2,N.3, we have

$$\begin{aligned} \frac{d[\text{N}]}{dt} &= k_{1f}[\text{O}][\text{N}_2] - k_{1r}[\text{NO}][\text{N}] - k_{2f}[\text{N}][\text{O}_2] + k_{2r}[\text{NO}][\text{O}] - k_{3f}[\text{N}][\text{OH}] + k_{3r}[\text{NO}][\text{H}] = 0 \\ \Rightarrow [\text{N}]_{\text{ss}} &= \frac{k_{1f}[\text{O}][\text{N}_2] + k_{2r}[\text{NO}][\text{O}] + k_{3r}[\text{NO}][\text{H}]}{k_{1r}[\text{NO}] + k_{2f}[\text{O}_2] + k_{3f}[\text{OH}]} \end{aligned} \quad (16)$$

The reaction rate constants, in $[\text{m}^3/\text{kmol s}]$, for 1-3 are as follows:

$$\begin{aligned} k_{1f} &= 1.8 \cdot 10^{11} \exp(-38,370/T) \\ k_{1r} &= 3.8 \cdot 10^{10} \exp(-425/T) \\ k_{2f} &= 1.8 \cdot 10^7 T \exp(-4680/T) \\ k_{2r} &= 3.8 \cdot 10^6 T \exp(-20,820/T) \\ k_{3f} &= 7.1 \cdot 10^{10} \exp(-450/T) \\ k_{3r} &= 1.7 \cdot 10^{11} \exp(-24,560/T) \end{aligned} \quad (17)$$

N_2O -intermediate mechanism is important in very-lean combustion ($\Phi < 0.8$). This mechanism can be represented by:



This mechanism is important in NO control strategies in lean-premixed gas turbine combustion applications. It has been shown that some NO is rapidly produced in the flame zone long before there would be time to form NO by the thermal mechanism. This is also known as the Fenimore mechanism. The general scheme is that hydrocarbon radicals form CN and HCN



The conversion of hydrogen cyanide, HCN, to form NO is as follows



For equivalence ratios higher than 1.2, chemistry becomes more complex and it couples with the thermal mechanism.